

Introduction to Reinforcement Learning

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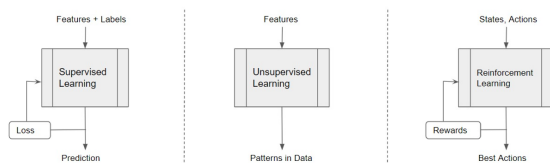
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What Is Reinforcement Learning (RL)?

- Reinforcement learning is a **goal-directed** computational approach where a computer learns to perform a task by interacting with a dynamic environment.
- This learning approach enables a computer to make decisions to maximize the cumulative reward for the task without **human intervention** and without being **explicitly programmed** to achieve the task. Examples include
 - Operating a drone or autonomous vehicle
 - Manipulating a robot to navigate the environment and perform various tasks
 - Managing an investment portfolio and taking trading decisions
 - Playing games such as Go, Chess, video games

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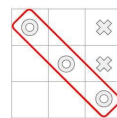
Comparison Between Supervised, Unsupervised and Reinforcement Learning



- Supervised Learning receives feedback from a Loss function acting as a 'supervisor'.
- Unsupervised Learning detects hidden patterns and receives no feedback from a supervisor.
- Reinforcement Learning gathers inputs and receives feedback by interacting with the environment and outputs the best actions

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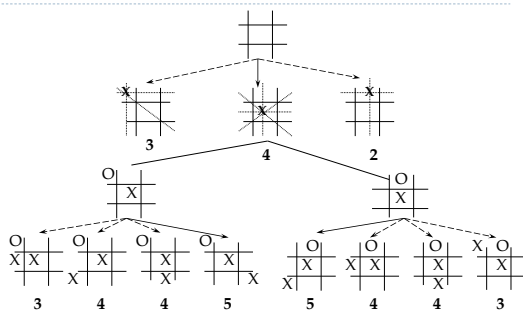
Inspiration from the Game of Tic-Tac-Toe



- Search exhaustively with 9! space
- Simple heuristic eliminates search entirely
- Heuristic: move to a position where X or O most winning lines
- Reasons for heuristics
 - impossible for exact solution,
 - heuristics lead to promising path
 - no exact solution
 - but an acceptable one
- Intelligence for a system with limited processing resources to make wise choices of what to do next

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Illustration of Heuristics in Tic-Tac-Toe



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Reinforcement Learning through Trial and Error

- With RL the learning happens from experience by trial and error, similar to a human, for example

- A baby can touch fire or milk and then learns from negative or positive reinforcement.
- The baby takes some actions
- Receives feedback from the environment about the result of the actions



- Repeats this process till it learns which actions produce favorable results and which actions produce unfavorable results.

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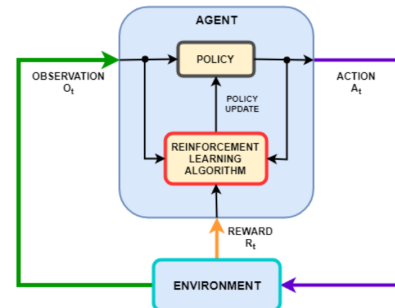
Agent

- ▶ The goal of reinforcement learning is to train an agent to complete a task within an unknown environment.
- ▶ The agent receives observations and a reward from the environment and sends actions to the environment.
 - ✓ The reward is a measure of how successful an action is with respect to completing the task.
- ▶ The agent contains two components: a policy and a learning algorithm
 - ✓ The policy is a mapping that selects actions based on the observations from the environment.
 - ✓ The learning algorithm continuously updates the policy parameters based on the actions, observations, and reward in order to maximize the cumulative reward received during the task.
- ▶ In other words, an agent learns the optimal behavior through repeated trial-and-error interactions with the environment without **human involvement**.

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Representation of Reinforcement Learning



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Reinforcement Learning Workflow

- ▶ Formulate problem – define the task for the agent to learn, including how the agent interacts with the environment and any goals the agent must achieve.
- ▶ Create environment – define the environment within which the agent interacts, including the interface between agent and environment.
- ▶ Define reward – specify the reward signal that the agent uses to measure its performance against the tasks and how to calculate this signal from the environment.
- ▶ Create agent – create the agent, which includes defining a policy representation and configuring the agent learning algorithm.

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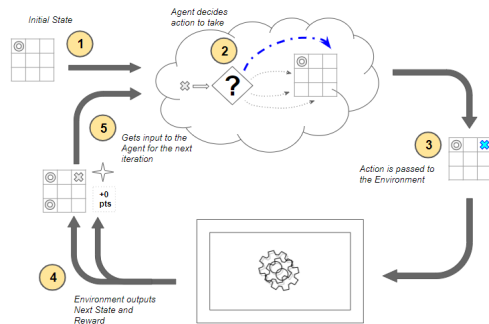
Reinforcement Learning Workflow (cont'd)

- ▶ Train agent – train the agent policy representation using the defined environment, reward, and agent learning algorithm.
- ▶ Validate agent – evaluate the performance of the trained agent by simulating the agent and environment together.
- ▶ Deploy policy – deploy the trained policy representation using, for example, generated GPU code.

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Formulate the Tic-Tac-Toe by Reinforcement Learning



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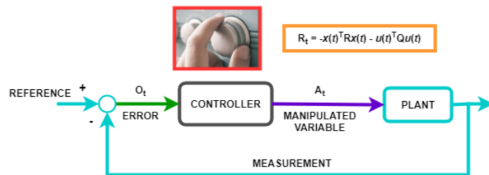
Formulate the Tic-Tac-Toe by Reinforcement Learning (cont'd)

1. The environment's current state is an input to the agent.
2. The agent uses that current state to decide what action it should take. It does not need a memory of the full history of states and actions that came before it.
3. That action is passed as an input to the environment.
4. The environment uses the current state and the selected action, and outputs two things – it transitions the world to the next state, and it provides some reward.
5. This reward from the environment is then provided as feedback to the agent as a consequence of the previous action.

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Reinforcement Learning for Control Systems



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Model Predictive Control (MPC)

- ▶ Many control problems encountered in areas such as automated plant, robotics and automated driving require complex, nonlinear control architectures.
- ▶ Techniques such as gain scheduling, robust control, and nonlinear model predictive control can be used for these problems, but often require significant domain expertise from the control engineer.
 - ▶ For example, gains and parameters are difficult to tune.
- ▶ The resulting controllers can pose implementation challenges, such as the computational intensity of nonlinear MPC.

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Mapping between RL and Control Systems

- ▶ **RL Environment:** everything that is not the controller. As seen previously, the environment includes the plant, the reference signal, and the calculation of the error. The environment can also include additional elements, such as:
 - ✓ Measurement noise
 - ✓ Disturbance signals
 - ✓ Filters
 - ✓ Analog-to-digital
- ▶ **Observation:** any measurable value from the environment that is visible to the agent. The controller can see the error signal from the environment. You can also create agents that observe, for example, the reference signal, measurement signal, and measurement signal rate of change and digital-to-analog converters

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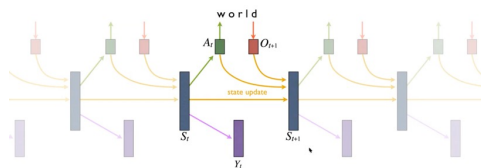
Mapping between RL and Control Systems (cont'd)

- ▶ **Action:** manipulated variables or control actions
- ▶ **Reward:** function of the measurement, error signal, or some other performance metric
 - ✓ For example, you can implement reward functions that minimize the steady-state error while minimizing control effort.
 - ✓ When control specifications such as cost and constraint functions are available, you can use to generate a reward function from an MPC object or model verification blocks.
 - ✓ You can then use the generated reward function as a starting point for reward design, for example by changing the weights or penalty functions.
- ▶ **Learning algorithm:** adaptation mechanism of an adaptive controller

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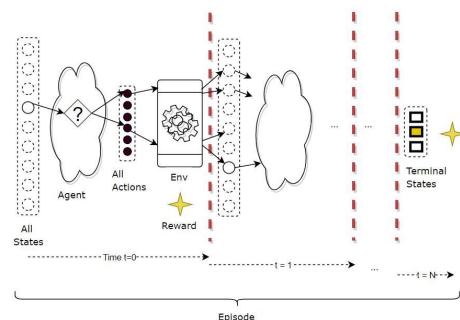
Formulate the Problem as Markov Decision Process

- ▶ Let's see how to use RL to solve a problem like game, controlled plant, etc?
- ▶ To apply RL, the first step is to formulate the steps of the problem solving as a decision making process as Markov Decision Process (MDP).



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Another View of Markov Decision Process



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Time-step and Episode

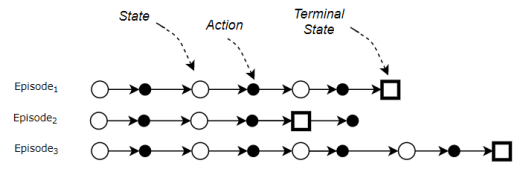
- ▶ In each time-step, three things occur — state, action and reward, which fully describe what happened in that time-step.
- ▶ For RL tasks that have a well-defined End or Terminal state, a complete sequence from the starting state to the end state is called an episode.
 - ▶ Each round of a game is an episode.
- ▶ So at the end of an episode, you can reset to a starting state (or randomly pick one from a set of starting states) and play another complete episode, and repeat.
- ▶ Each episode is independent of the next one.
- ▶ Therefore an RL system's operation repeats over multiple episodes. Within each episode, it repeats over multiple time-steps.

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Time-step and Episode (cont'd)

- ▶ The execution of the MDP can be described as a trajectory of occurrences (in terms of state, action, reward) over a sequence of time-steps within episode as follows: $(s_3, a_3, r_4, s_4, a_4, r_5, s_5, a_5, r_6, s_6, \dots)$

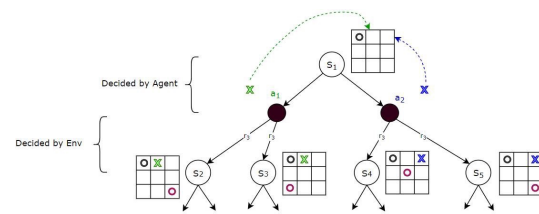


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Mapping MDP to Tic-Tac-Toe

- ▶ Given the current state, the next action is decided by the agent. In fact, that is the only job of the agent. For instance, from the current state, the agent can choose, say, action a_1 or a_2 to place its token.



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